SENIOR QA ENGINEER · SOFTWARE QUALITY, AUTOMATION & RCA

William Lu

Senior QA Engineer | Software Quality Assurance, Test Automation & Root-Cause Analysis

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LinkedIn | QA Portfolio | GitHub

★ DOMAIN	FINTECH QA	LEADERSHIP	IMPACT
17+Y Tech	@BTSE	@ASML	@TrendMicro
12+ Yrs QA	**3 High-Sev**	**Led 6 Eng.**	**-10% · -20pp**
Security, Semi, FinTech · 2× Best SEG Engineer	Futures balance, Spot zero-fee & Convert lock bugs	Remote HMI eP5 TestComplete + Bamboo CI	DLP CPU/RAM cut & 20pp fewer L2 case returns

Professional Summary: Senior QA Engineer with **12+ years in software quality assurance and escalation engineering** (within 17+ years in technology) across **Enterprise Cybersecurity, Semiconductor Automation, FinTech/Exchange Platforms, and Embedded Systems**. Proven track record leading a 6-engineer remote automation team (**ASML**), uncovering high-severity financial-logic defects via API/WebSocket/SQL validation (**BTSE**), and driving deep system/dump RCA and performance optimization (**Trend Micro**).

CORE SKILLS & TECHNICAL STACK

Test Strategy & QA Leadership
Risk-based Testing, Test Specifications & Test Data Design, Test Reviews, QA Leadership, Internal/External Beta Planning, Automation Feasibility Assessment, Reusable-Module Planning, Release Sign-off & Handover

Functional, Acceptance & System Testing
Manual & Exploratory Testing, Functional & Regression Testing, Boundary / Negative / FET (Force-Error Test), RAT / FAST / TOFT, System & Compatibility Testing, Cross-browser (Chrome, Edge, Firefox), Performance, Load & Stress Testing

Automation, API & Data Validation
Python, Selenium, Robot Framework, TMSTAF, TestComplete, Jenkins, Atlassian Bamboo; E2E Web API Validation, WebSocket Event Inspection, Browser DevTools, SQL, TablePlus, Excel Formula Reconciliation

Troubleshooting, Environments & Tools
Defect Triage, Customer-Escalation Reproduction, Incident RCA, Hotfix Validation, Log & Memory Dump Analysis (`WinDbg` , `ProcMon` , `Process Explorer` , `Event Log`); TestRail, Jira, JQL, Confluence, GitHub, Bitbucket; VMware (Workstation, vSphere/ESXi), VirtualBox, Hyper-V, Windows Server, Azure

2025–Present Self-Directed Practice & Product Domains
2025–Present Projects: Python, `pytest` , `Playwright` , `GitHub Actions` | **Domains:** Enterprise Security & DLP SDK; Financial & Exchange Workflows (Wallet, Spot, Futures, Convert, Referral/Affiliate, Fees, Funding, Liquidation, KYC)

PROFESSIONAL EXPERIENCE

Career Break | Family Caregiving

Feb 2025–Present

- Planned career break for family caregiving; maintained hands-on QA engineering practice through public software quality projects using Python, `pytest` , `Playwright` , and `GitHub Actions` .

Nogle (BTSE) | Senior QA Engineer

Sep 2024–Nov 2024

Taipei, Taiwan · On-site

→ Full Case Study

- **Escalated 3 high-severity financial-logic defects** via boundary, negative, API, WebSocket, and SQL validation: **(1)** Futures balance miscalculation after partial close and funding settlement (un-updated DB state), **(2)** Spot zero-fee rounding defect on micro-orders, and **(3)** Referral rebate precision loss alongside a Convert expired-quote rapid-click lock defect.
- **End-to-End Exchange Coverage:** Authored test specs and **50+ TestRail/Wiki test cases** (`30-50` Referral/Affiliate, `~20` Convert); validated Wallet, deposit/withdrawal, blockchain confirmation, Spot, Futures, Convert, maker/taker fees, funding settlement, liquidation, and Individual/Corporate KYC.
- **Multi-Layer Data Reconciliation:** Verified real-time consistency across UI, REST APIs, WebSocket streams (`executionReport` , `orderUpdate` , `outboundAccountPosition` , `positionUpdate` , `Balance Update`), TablePlus SQL queries, and Excel formula models.

STACK: `REST API · WebSocket · Browser DevTools · SQL (TablePlus) · TestRail · Excel Formula Reconciliation · Chrome/Edge/Firefox`

Zealogics | Project Leader / Senior Automation Test Engineer

May 2023–Apr 2024

Fully Remote · Client Engagement: ASML

→ Full Case Study

- **Remote Team & Delivery Leadership:** Led **6 Automation Engineers** in a fully remote ASML engagement covering HMI eP5 modules (`iCDU` , `MM Flow` , `ADEL`), managing weekly progress via custom Jira Filters/JQL, code/test reviews, annual performance evaluations, and client TOI/live demos.
- **Unattended CI Lifecycle Design:** Architected the end-to-end test execution flow (Simulator startup, initialization, Recipe loading, execution, reporting, cleanup, and environment reset) within TestComplete—enabling **unattended normal-path execution** via `.bat` and Atlassian Bamboo triggers.
- **Modular Reuse & Feasibility:** Assessed manual test cases for full vs. partial automation, planned shared modules prior to assignment to eliminate duplicate coding, authored Test System Design docs in Confluence, and contributed direct test implementation alongside the team.

STACK: `Python · TestComplete · Atlassian Bamboo · Bitbucket · VS Code · Jira (JQL) · Confluence`

Trend Micro

May 2010–Jan 2023

Taipei, Taiwan · On-site

→ Full Case Study

Senior QA Engineer & Senior Customer Service Engineer

Jul 2014–Jan 2023

- **QA & SEG Leadership:** Served as **QA Leader and Service Engineer Group (SEG) Leader** for the Data Loss Prevention (DLP) SDK and enterprise security products (`OfficeScan` , `WFBS-SVC` , `IMSVA`), bridging system/performance QA with global customer-escalation RCA and hotfix validation.
- **Proactive Chrome Compatibility Automation:** Built Python monitoring via Google APIs (`googleapis.com`) to detect Chrome Beta/Stable updates, auto-run sensitive-file upload tests, and send alerts when DLP blocking failed—catching browser breaks before customer impact.
- **Performance Optimization (-10% CPU/RAM):** Proposed **supported-site scanning** during a DLP refactor; QA benchmarks confirmed **CPU and memory usage each fell by at least 10%** compared to the prior architecture.
- **Support Diagnostic Standardization (-20pp Returns):** Initiated an issue-type-based DLP diagnostic guide; in its first month, **cases SEG returned to L2 for missing evidence dropped by 20 percentage points** (per manager metrics).
- **Deep Escalation RCA & Recognition:** Automated regression suites and reproduced complex customer crashes/leaks across multi-hypervisor labs using memory dump and kernel/process diagnostics; named **Yearly Best SEG Engineer in 2019 and 2021**.

STACK: `Python · Selenium · Robot Framework · TMSTAF · Jenkins · WinDbg · ProcMon · Process Explorer · VMware · Hyper-V · Azure`

Senior Customer Service Engineer

Jun 2013–Jul 2014

- Served as SEG Leader and final escalation point for critical WFBS Advanced incidents across APAC, Europe, and the US; led dump/upgrade-log RCA, reconstructed customer-upgrade-path hotfix validation, and proposed the WFBS Guide Book to reduce missing-info case returns.

QA Engineer

May 2010–Jun 2013

- Owned module, system, and long-running performance testing (Python + Windows Performance Monitor) for WFBS Advanced (`6.0 SP1–7.0 SP1`); served as **Internal & External Beta Owner**, leading on-site production beta deployments and field diagnostics across Europe.

Tatung | Firmware Engineer

Apr 2006–Apr 2010

Taipei, Taiwan · On-site

- Developed and validated embedded C firmware for European/US LCD TVs (from-scratch OSD UI and A/V hardware calibration) and Java control systems for smart meters, supporting stable mass-production releases and a GE OEM engagement.

SELECTED QA PROJECTS & ENGINEERING CASE STUDIES (2025–PRESENT)

Market Data Quality Lab ↗

Python & `pytest` validation suites checking REST snapshots, WebSocket sequencing, latency boundaries, and order-book invariants on public market feeds.

QA Portfolio & Risk-Based QA Decision Desk ↗

Bilingual QA portfolio validated by Python, `pytest` , `Playwright` , and `GitHub Actions` across static, runtime, responsive, and accessibility checks.

BTSE Financial-Logic Case Study

ASML Automation Leadership | Trend Micro RCA & Beta

Trend Micro DLP Compatibility & Perf.

EDUCATION

MS, Information Engineering & CS — Feng Chia University 2003–2005

BS, Information Engineering & CS — Feng Chia University 1998–2003

WORK PREFERENCES & AVAILABILITY

Based in **Taiwan (UTC+8)**; targeting **Fully Remote** roles with **4+ hours of European business-hour overlap** (`09:00–24:00` Taiwan time) and up to **25% travel**. Languages: **Mandarin** (Native), **English** (Professional working proficiency).